

SYED MAOZ ARIF

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SUMMARY

Detail-oriented Hardware Developer and Electronics Engineer with a proven track record in multi-layer PCB design, embedded systems, and IoT architecture. Developed 12+ real-world engineering projects and achieved 17 technical certifications. Recognized for innovative problem-solving, securing First Position in national engineering exhibitions and a Global Instructables Contest win. Adept at cross-functional leadership, managing 50+ members serving as IET Chapter President, and delivering high-quality freelance engineering solutions with a 100% client satisfaction rate.

EDUCATION

COMSATS University Islamabad, Abbottabad Campus

B.S. Electrical Engineering (Electronics)

Expected Aug 2028

Abbottabad, Pakistan

- **Honors:** Altium Global Scholarship Awardee (Selected among top 1% globally for engineering excellence).
- **Relevant Coursework:** Embedded Systems, Digital Logic Design, Signal Integrity, C/C++ Programming.

WORK EXPERIENCE

Freelance Hardware & Mechanical Designer

Fiverr & Upwork

Jan 2024 – Present

Remote

- Deliver over 15 multi-layer PCBs and 3D CAD models to global clients, achieving 100% on-time delivery and 5-star ratings across four industry sectors.
- Optimize component selection (BOM) and schematic captures, reducing overall hardware prototyping costs by 25% for 10+ clients.

President (Previously VP & Membership Head)

IET On Campus, CUI Abbottabad

May 2025 – Present

Abbottabad, Pakistan

- Direct chapter operations and manage a 5-person cross-functional executive team, fostering engineering excellence for 50+ active members.
- Organize 5+ technical workshops (e.g., AI in UAVs) with 100+ attendees, increasing year-over-year student engagement by 40%.

PROJECTS

Industrial Edge AI Condition Monitoring & 3D Digital Twin

Jul 2026

- Designed a dual-MCU industrial development board integrating 70 GPIOs, reducing 3-phase induction motor diagnostic time by 40% for predictive maintenance applications.
- Secured First Position (Software Category) at CEP Exhibition SP26 and 2nd Position at EMCOT 2026 out of 50+ competing projects.

Stratum-1 Precision Hardware Clock & GPS Bridge

Aug 2026

- Engineered a hardware time synchronizer using pure digital logic, bypassing OS latency to achieve 0 ms delay and 100% synchronization accuracy.
- Developed a custom Verilog UART receiver and NMEA state machine on a Lattice FPGA to process live GPS signals directly in silicon.

ElectroParts IMS: Inventory Architecture

Aug 2026

- Architected a cross-platform inventory management system supporting 32 distinct electronics categories, increasing component retrieval efficiency by 50%.
- Implemented dual-storage local/cloud synchronization and linear barcode generation, enabling zero-latency data access for 1000+ logged components.

InclSense Precision Digital Level

Apr 2026

- Built a precision leveling device analyzing dual-axis spatial data via Inter-Integrated Circuit (I2C), yielding high-accuracy PC telemetry for real-time monitoring.
- Awarded Global Runner-Up in the Instructables Sensors Contest out of 500+ international submissions.

TECHNICAL SKILLS

Hardware Design: Altium Designer, KiCad, Proteus, LTspice, Multisim, Schematic Capture, Signal Integrity
Embedded Systems: ESP32, STM32, Arduino, Lattice FPGA, I2C, SPI, UART, MQTT, Firebase
Programming: C/C++, Python, Verilog (HDL), MATLAB, JavaScript, HTML5/CSS3
3D CAD & Tools: SolidWorks, Fusion 360, AutoCAD, 3D Printing, Git, LaTeX, Agile Methodologies

CERTIFICATIONS

Altium Certified: Designer Essentials <i>Altium Education</i>	<i>Mar 2026</i>
Generative AI Mastermind <i>Outskill</i>	<i>Jan 2026</i>
Arduino & IoT Certification <i>Cardiff Institute</i>	<i>Aug 2025</i>
3D Printing & CAD Engineering Certificate <i>Online Institute</i>	<i>May 2025</i>